



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End Spring Semester Examination 2025-26

Sub No: CS60216 Sub Name: Safety fundamentals for Generative AI
Department/Centre/School : Computer Science and Engineering
Duration 3 hrs Marks = 80

Specific charts, graph paper, log book, etc. required NO

Special Instructions: ANSWER ALL questions. All parts of a single question should be answered together. Answers should be brief and to-the-point. Marks will be deducted for sketchy answers and claims without proper reasoning. **No clarifications!** In case of reasonable doubt, make assumptions and state them upfront. You can keep probability values in fractional forms.

1. RESTA

- (a) If a PEFT model has an unsafety score of 35.27% and RESTA reduces it to 13.59%, calculate
i. percentage reduction
ii. if this reduction trend continues, predict unsafety after one more equal reduction cycle. [1+2=3]
- (b) Given two orthogonal vectors: $\|\delta_{safe}\| = 3.5$ and $\|\delta_{task}\| = 4.2$ and the final net δ after applying RESTA is $\delta_{net} = \delta_{task} + (\gamma - \lambda)\delta_{safe}$ with $\gamma = 1.5$ and $\lambda = 1.0$. Compute the following.
i. The total net norm $\|\delta_{net}\|$. 4.55
ii. The angle between δ_{net} and δ_{safe} . 67° [2+3=5]
- (c) Prove that if two safety vectors δ_{safe}^1 and δ_{safe}^2 are already aligned in the vector space then combining them via RESTA is equivalent to applying RESTA once with their sum. [4]

2. Model Arithmetic

- (a) Consider the paradigm of Model Arithmetic to answer this question. For union operation over three models with tokens x and y we have the following: $\log P_1(x) = -1.8$, $\log P_2(x) = -2.5$ and $\log P_3(x) = -1.9$. Corresponding log-probs for y are all -0.3 . Compute
i. Union log-prob for (i) x and (ii) y . -1.2 -0.3
ii. Final normalized probability for x and y after passing through softmax. [1+1+2=4]
- (b) Consider the paradigm of Model Arithmetic to answer this question.
i. Prove that the linear combination of log-probs of two models minimizes the sum of weighted KL divergences to each model. In other words show that $P^*(x) \propto \exp(\lambda \log P_1(x) + (1 - \lambda) \log P_2(x))$ minimizes $L(P) = \lambda D_{KL}(P||P_1) + (1 - \lambda) D_{KL}(P||P_2)$.
ii. Now consider $\log P_1(x) = -1.5$, $\log P_2(x) = -2.5$ Find λ that minimizes KL divergence of interpolated model to target log-prob: $\log P^*(x) = -2.0$.

[6+3=9]

3. Corrigibility

- (a) A casino offers the following: (a) \$2000 if you win both Game A and Game B (b) \$500 if you win only one of them (c) \$0 if neither. Probabilities: (i) Game A win: 40% (ii) Game B win: 30%. Compute total expected value. *470*
- (b) A high-stakes roulette offers: \$10,000 with $p = 0.01$ and \$0 with $p = 0.99$. You may increase p to 0.015 for a fee of \$80. Is it worth paying? Why or why not? *100, 10 100*
- (c) Suppose $u_B = 100$, $u_N = 90$. If the AI agent can increase $P(B)$ from 0.3 to 0.4 by reducing u_N by 5, should it do so? *97, 91 100*

[3+2+3=8]

4. Scalable Oversight

- (a) In a task with debate protocol, judges select correct answers 70% of the time when the protagonist picks correctly, and 50% when they pick incorrectly.
- Compute expected judge accuracy if the protagonist picks correctly in 80% of cases.
 - How does accuracy shift if protagonist correctness drops to 60%?
- (b) Given the effort per response = $\epsilon(R)$. Effort for different critique levels are $C_1 = 3\epsilon$, $C_2 = 5\epsilon$ and $C_3 = 7\epsilon$.
- How many responses would equal the computational effort of a C_3 ?
 - If response accuracy is $57\frac{1}{7}\%$, what is the expected majority-vote accuracy (from 7 responses)?

[2+2=4]

[1+2=3]

5. Low-rank Adaptation, Quantization

- (a) Suppose that you want to use Low-rank adaptation methods to train the GPT-2 model, which has a model dimensionality of 1600, 48 decoder layers and 25 attention heads, for the tasks of text simplification. Consider the following scenarios. What will be the number of parameters to be trained? What will be the number of parameters at the inference time?
- You are using LoRA only for the query and key matrices, each with a rank of 8.
 - You are using 2 adapters per layer, each with a down projection of 8.
- (b) Suppose that you are using QLoRA to achieve 2-bit quantization with values: $\{-1, -0.3, 0.5, 1\}$. Assume your input tensor is $[38, -3, -50, 25]$. What will you need to store in the quantized format. Also, what will be the output after dequantization? *[3]*
- (c) Suppose that you are using double quantization for a 10×10 weight matrix containing FP32 values to store as Int8. You are using a block size of 20 for the double quantization. How many bits would you save? *2320*

[3+3=6]

[3]

[3]

6. Mechanistic Interpretability

(a) While viewing feed-forward networks as key-value memories, how many distinct memory coefficients would be there in a standard GPT architecture, with 768 as the model dimension and 12 layers? How many distinct key and value vectors? Now, consider a key and value vector at a given layer. Explain briefly how to identify the pattern associated with the key, and the vocabulary distribution for the value. [5]

(b) Explain the formulation of Tuned lens. How does it improve upon the original idea of Logit Lens? [2]

(c) What are the roles of QK circuit and OV circuit in the residual stream of a token? [2]

7. Causal Tracing

(a) Suppose that you are running a causal mediation analysis over GPT-2, which has a model dimensionality of 1600, 48 decoder layers and 25 attention heads. Also, suppose that the third to fifth tokens out of twelve tokens in the input sentence belong to the subject.

i. In the corrupted run, how many states of the model (as defined, include the embedding state as well) will get corrupted?

ii. Also, suppose that you are allowed to restore one of these corrupted states. Which one would you restore to have the maximum impact on the output prediction for the sentence? [2+1 = 3]

(b) Suppose that you want to edit the MLP memory of GPT-2 using ROME method, so that it starts associating 'Eiffel Tower' with 'Delhi'. You also know that it is better to modify the 24th layer for the same.

i. Which parameters of the FFN will you modify? The up-projection or down-projection? How many parameters of FFN are guaranteed to be the same as before?

ii. What is the exact update to the matrix? Provide the formulation. How would you choose the various terms required for the update? You can discuss any two of the terms. [3+3 = 6]

8. Watermarking

(a) Suppose that you are using hard red list for watermarking, with equal division of the vocabulary into 'green list' and 'red list'. Suppose your z threshold is 3.

i. What is the minimum length of text for which you can detect watermarking?

ii. Suppose that you have a watermarked sequence of length $T = 1000$ tokens. Suppose that an adversary modified 220 of these tokens. In the most adversarial scenario, can you still detect the watermarking? If not, what is the minimum number of tokens that the adversary will have to modify to bypass the watermarking? 227

(b) Suppose now that you are using a soft red list for watermarking with $\gamma = 0.6$ (green list is of size $\gamma|V|$), and $\delta = 0.5$. [2+3 = 5]

i. Suppose that you are having a sequence of length $T = 100$ tokens, out of which you detected 72 green words. What is the z -score?

ii. Assume your vocabulary is of size 5. At a given point in time, show an example logit score distribution for the vocabulary, such that the watermarking scheme may not be able to enforce a green token? Also, give another distribution for which the scheme will surely be able to enforce the green token. Avoid uniform distributions. [3+2 = 5]